ALGEBRA LADDER 4 OF 9

Letters as numbers: expanding, factorising and proof

A letter stands for any number, so a line of algebra makes a claim about every number at once.

1 / THE RULE AND THE PICTURE

To show a claim holds for every number, write the number with a letter and use the laws



$5^2 = 25$ teal dots
$+ 11$ gold dots $= 6^2 = 36$

Going from a square of n dots a side to a square of $n+1$ adds an L-shaped strip of n dots down one side, n along the other and one in the corner, and the picture holds for every n:

$$\boxed{(n+1)^2 = n^2 + 2n + 1}$$

Expanding $(n+1)(n+1)$ with the distributive law gives the same three terms, so the dots and the algebra agree, and the algebra covers numbers too large to draw.

2 / EVEN AND ODD NUMBERS

Every even number is $2m$ and every odd number is $2m+1$

Two odd numbers add to $(2m+1)+(2n+1) = 2m+2n+2 = 2(m+n+1)$, which has a factor 2 and so is even. Two odd numbers multiply to $(2m+1)(2n+1) = 4mn+2m+2n+1 = 2(2mn+m+n)+1$, which is odd.
Each step uses only the commutative, associative and distributive laws, and each holds for every pair of odd numbers, which no list of examples could show.

3 / WHY A NUMBER TRICK WORKS

Follow the letter through the steps

step	in letters	with $n = 7$
think of a number	n	7
add 4	$n+4$	11
double it	$2n+8$	22
subtract 8	$2n$	14
halve it	n	7

Whatever number n is chosen, the steps $n \to n+4 \to 2n+8 \to 2n \to n$ return it, so the trick "think of a number, add 4, double, subtract 8, halve" always gives back the starting number. The doubling used distributivity, $2(n+4) = 2n+8$, and the halving used the inverse of doubling, so the trick is the undo machine of the previous sheet hidden behind a story.

4 / A PRODUCT FROM A DIFFERENCE

$a^2 - b^2 = (a-b)(a+b)$ turns squares into quick sums

Expanding the right side gives $a^2 + ab - ba - b^2 = a^2 - b^2$, since $ab = ba$.
This makes $51^2 - 49^2 = (51-49)(51+49) = 2 \times 100 = 200$ and $99^2 - 1 = 98 \times 100 = 9800$ immediate.
Taking $a = n+1$ and $b = n$ shows that $(n+1)^2 - n^2 = 2n+1$, so each new square adds an odd number of dots, as the gold strip showed: $8^2 - 7^2 = 15$.

5 / CHECKING IS DIFFERENT FROM PROVING

A pattern that holds many times can still fail

The expression $n^2 + n + 41$ gives a prime for every n from 0 to 39, yet at $n = 40$ it gives $40^2 + 40 + 41 = 1681 = 41^2$. Substituting numbers is a good way to catch a mistake, such as testing $(x+3)(x+2) = x^2 + 5x + 6$ at $x = 4$, and a proof with letters is what settles a claim for all numbers.

6 / TRY THESE

Use a letter and a law

(a) Show that three consecutive numbers add to a multiple of 3: $n + (n+1) + (n+2) = 3n + 3 = 3(n+1)$, as in $7 + 8 + 9 = 24 = 3 \times 8$.
(b) Show that the sum of an even and an odd number is odd: $2m + (2n+1) = 2(m+n) + 1$.
(c) Use $a^2 - b^2 = (a-b)(a+b)$ to find $1001^2 - 999^2 = 2 \times 2000 = 4000$.

ASIDE / THE POLYNOMIAL THAT FAILS

The expression $n^2 + n + 41$ is Euler's prime-generating polynomial, and the hexagonal spiral sheet explains why its run stops at 40.

LINKS TO OTHER SHEETS

The next sheet turns to operations that do not commute; the number theory ladder uses the same letters to prove statements about divisors and remainders.

Convention. n, m, a, b stand for integers; even is $2m$ and odd is $2m+1$; a proof uses letters, a check uses numbers.